Engineered Intelligence: Making Physical Design Executable

Executable Genomes, Recorded Inheritance, and Full Population Evidence

Research draft prepared for Jepson Taylor | Human author approval pending

Abstract—Engineered Intelligence treats physical design as an executable process: a genome defines geometry, numerical models evaluate it, and an automated population search explores alternatives. We present Jarambit, a Rust-centered research system connecting a 128-coordinate geometric genome and categorical material state to CAD decoding, island-based evolutionary search, and multiple finite-element assessment models. Its retained records include 1,954,560 evaluations across 10 four-island runs spanning generations 0–800 (Run 2–Run 11, 195,456 births per run, 754,236 search-legal), with complete compact genome and score tables, alongside a reconstructable historical record (Run 1, 1,984 births across generations 0–16) that exposes parentage, 2,176 island-slot memberships, 192 elite reappearances, and 12 inter-island migrations, plus 3,649 intermediate multi-stage screening proposals (2,259 joined to Stage-1 observations and 1,390 unscreened). Source-bound population views show the complete retained search record while keeping different scorers and assessment stages separate. The evidence also exposes a critical limitation: a high search score does not establish later mechanical qualification; the retained historical top-ranked birth (Individual 1095, $f = 1.15727$) and the single assessed automatic-pipeline export (Individual 2000; 1 assessed candidate, 0 qualified) fail strict verification. Matched-mesh quadratic solid solver agreement ($< 0.05\%$) and refused plate grid-convergence estimates provide complementary checks rather than proof of physical performance. The contribution is an inspectable connection between representation, search, and engineering evidence.

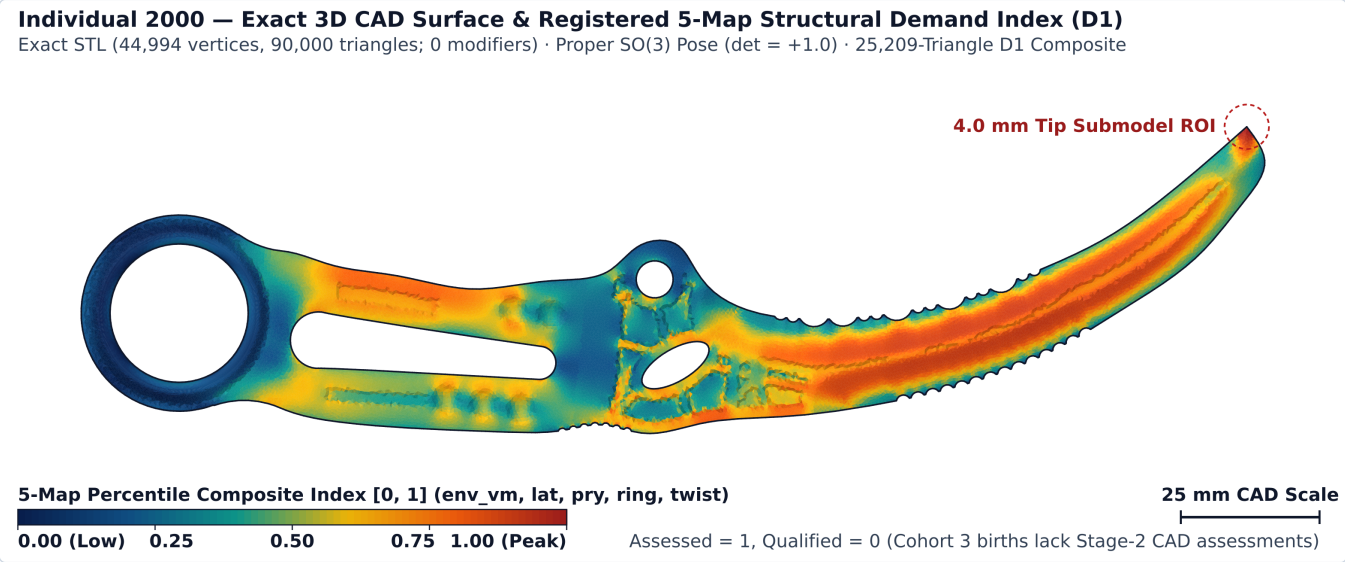


Fig. 1. Illustrated assessed export Individual 2000 (single assessed candidate in the automatic certification leaderboard, 1 assessed, 0 qualified, no promoted champion; separate from the 23-subject historical battery, while the 1,954,560 native search births lack retained Stage-2 CAD assessments). Full-width white-background exact-CAD 3D surface projection (44,994 vertices, 90,000 triangles, zero modifiers; proper $\mathrm{SO}(3)$ pose $\det = +1.0$, dark silhouette and hole contours) colored by the dimensionless 5-map percentile composite index in $[0, 1]$ across the retained $0.25\,\mathrm{mm}^2$ plate mesh (25,209 triangles, 13,503 nodes: multi-case envelope plus unit-response lateral, pry, ring-traction, and torsion maps; in the envelope, both lateral and pry unit maps are scaled by $260\,\mathrm{N}$ alongside $1500\,\mathrm{N}$ ring traction and $1.82\,\mathrm{N}\cdot\mathrm{m}$ torsion). Detailed provenance, raw Cycles history, and unlit reference plates appear in Sections V–X.

Index Terms—Engineered Intelligence, computational design, evolutionary computation, executable genomes, CAD, quality diversity, finite elements, evidence provenance, reproducible populations.

IEEE-style research white paper, not an accepted IEEE venue article. Retained computational and software records only; no new CAD/FEM search or physical experiment was executed for this manuscript. Each experimental cohort is bound to its specific producer revision and archival provenance in Section X, separating reconstructed historical search, multi-stage screening, and the ten-run generation-800 population archive.

I. Introduction: Engineered Intelligence as an Executable Design Process

ENGINEERING design is traditionally documented through its terminal artifact: a finished CAD assembly, a selected stress contour, and a summary table asserting that requirements were met. When evolutionary computation and automated numerical modeling enter the loop, that artifact-only view conceals the primary scientific object. The primary object is no longer a single hand-tuned shape, but an executable design process that generates, evaluates, retains, and relates candidate geometries under explicit numerical preferences.

We define *Engineered Intelligence* operationally through five coupled components: a versioned numerical representation, a deterministic geometric decoder, a recorded variation and selection loop, declared stage-specific objectives, and an inspectable archive of every evaluated candidate and observation. Defining the discipline operationally avoids both overclaiming autonomous invention and allowing test-catalog administration to obscure how candidate geometries actually evolve.

A. The Executable-Representation Mechanism

Jarambit instantiates this architecture on a monolithic titanium karambit structural design problem. Every candidate is encoded by 128 continuous floating-point coordinates paired with one categorical alloy state (`Ti-6Al-4V` or `Ti-3Al-2.5V`), forming a 129-field genome record. Decoding a genome maps 40 normal spline offsets around a closed reference outline, global planform and ring dimensions, stock-to-tip taper and bevel angles, and conditional interior pocketing or continuous-rail thickness laws into a watertight B-rep solid, a surface mesh, and a 2.5D plate/membrane or 3D quadratic tetrahedral finite-element discretization. Within each evolutionary run, four subpopulations (Islands 1–4) apply block-preserving silhouette crossover and six structured mutation modes to generate offspring, evaluate their geometric legality and mechanical response, update a behavioral quality-diversity archive, and periodically migrate elites around an island ring.

B. Research Question and Three-Cohort Empirical Findings

The central research question of this study is: *How do representation constraints, contextual search objectives, and multi-fidelity numerical verifiers interact across complete evolutionary populations, and why must search-stage preference remain separated from engineering qualification?*

Because historical search runs, multi-stage screening waves, and long-horizon quality-diversity campaigns used distinct scorer revisions and retention policies, we analyze the empirical record across three separate cohorts without pooling incompatible scores into a synthetic grand total:

1) **Cohort 3 (10-Run 800-Gen Population, 1,954,560 Evaluations — Run 2 through Run 11):** Ten retained four-island runs completed 801 generations (0–800) with 195,456 evaluated births per run, providing 1,954,560 complete 146-column records (17 scalar evaluation metrics plus all 129 genome-and-material fields) and 32,040 generation-by-island summary rows. Across these ten retained runs, 754,236 evaluations (38.59%; per-run mean $75{,}423.6$, sample SD $2{,}936.55$, population SD $2{,}785.86$) satisfy search-level geometric legality, and run-maximum native quality-diversity fitness reaches between 1.1089 (Run 3) and 1.1440 (Run 5) (mean 1.1250, sample SD 0.01285, population SD 0.01220). An eleventh staged summary cohort (Run 12–Run 22, 17,672 generation-by-island summary rows reaching generation 708) records earlier summary-only trajectories.
2) **Cohort 1 (Historical 17-Gen Genealogy — Run 1):** A fully reconstructable 17-generation four-island experiment retains 1,984 births (1,455 with completed plate finite-element solves, 529 not evaluated by finite elements, and 446 search-legal births), 2,176 island-slot membership occurrences ($4 \times 32 \times 17$, including 192 elite reappearances), 128 seed births, 938 mutation births, 918 crossover births, and 12 ring-migration events at generations 4, 8, and 12. Tracing the three-hop ancestry of the highest-scoring birth (Individual 1095 at generation 9, native fitness 1.15727) reveals ancestor reuse across branches (Individual 413 and Individual 174) and shows that Individual 1095 fails subsequent strict regional yield-order sequence verification.
3) **Cohort 2 (Intermediate Multi-Stage Screening):** Across 24 multi-stage screening runs, 3,649 birth proposals are retained, of which 2,259 link via validated one-based candidate identity to Stage-1 screening scores and 1,390 remain in an explicit unscreened observation band rather than being zero-filled or conflated with solver failures. Within the six retained ultralight reporting curves (15 generation summaries), Individual 2002 records the highest retained weighted reporting profile score (74.207299 on the 0–100 scale, display loss 25.792701), while separately assessed controlled exports Individual 2001 ($W_{\mathrm{CAD}} = -3.878704$, $G_1 = 0.623903$, mass $75.382\,\mathrm{g}$, 11 of 16 legacy gates met and 5 failed) and Individual 2007 ($W_{\mathrm{CAD}} = -3.964924$, $G_1 = 0.621634$, mass $75.642\,\mathrm{g}$, 11 of 16 legacy gates met and 5 failed) fail strict multi-gate verification.

For detailed spatial and multi-solver mechanics, we examine the illustrated assessed geometry Individual 2000 (the single assessed export in the automatic certification leaderboard, 1 assessed candidate, 0 qualified, no promoted champion; separate from the 23-subject historical battery, while the $1{,}954{,}560$ native search births lack retained Stage-2 CAD assessments). Its retained $0.25\,\mathrm{mm}^2$ plate discretization ($25{,}209$ triangles, $13{,}503$ nodes across seven retained mesh arrays) and $91{,}008$-element quadratic tetrahedral tip submodel ($416{,}583$ displacement degrees of freedom) demonstrate 0.04%–0.05% cross-solver agreement between independent finite-element engines while exposing non-monotonic and negative-order plate grid-convergence behavior that must be refused rather than smoothed over.

C. Bounded Contributions and Provenance Scope

This paper makes four bounded contributions:

1) **Executable Genome and Variation Exposition (Sections III–IV):** We document the exact 129-field genome contract, decoder clamping couplings, block-preserving silhouette crossover, and six-mode mutation operators, linking them to recorded multi-hop genealogies and coordinate-level parent-child differences.
2) **Full-Population Archival Census Across Three Cohorts (Sections IV and VIII):** We present the complete 1,954,560-evaluation ten-run generation-800 population alongside the 1,984-birth historical genealogy and 3,649 intermediate screening proposals, maintaining strict separation between birth clouds, island memberships, and stage-specific score profiles.
3) **Source-Bound Multi-Model Mechanics and Saliency Visualization (Sections V–VII):** We define area-weighted midrank percentile composites across the five structural demand maps (multi-case von Mises envelope and unit lateral, pry, ring-traction, and torsion responses) and $\geq 90\%$ integrated-saliency local crops over the retained plate fields of Individual 2000, while moving the exhaustive 414-cell historical battery matrix and 44-entry technical identifier crosswalk to the Supplement.
4) **Compact Lossless Population Archive Specification (Section X):** We specify and verify a segmented zstd population archive format (200 segments compressing $3.81\,\mathrm{GB}$ of raw CSV records into $494.8\,\mathrm{MB}$ with 100% round-trip identity) that separates historical reconstruction revisions, observed repository checkouts, and executed binary provenance.

Scope and Non-Claims: All results reported here are retrospective computational records; each cohort is attributed to its specific producer revision in Section X rather than attributed to a single uniform binary. No candidate in any cohort is physically qualified, no hardware was machined or mechanically tested, and no claim is made of fleet-wide completion (10 of 18 native run directories reached generation 800) or autonomous representation design.

II. Background and Related Research

A. Executable CAD Representations

DeepCAD represents a shape as a sequence of CAD operations and learns a Transformer-based model for autoencoding and generation [2]. Text2CAD conditions sequential parametric CAD generation on natural-language descriptions and evaluates visual, parametric and geometric outcomes [3]. These works make construction history part of the generated representation. Jarambit's retained crescent model instead decodes a fixed numerical schema into geometry. It does not learn a general CAD-operation language, and this study supplies no common benchmark against either method.

This difference matters when interpreting a genome. More numerical parameters do not necessarily express a new topology or construction strategy. A decoder revision can change the represented family without changing vector length. Representation identity therefore includes schema and decoder, not just gene count. The prospective agent research in Section IX concerns assistance with that representation; it is distinct from the implemented search over its parameters.

TABLE I
Related methods and their reported evaluation scope. These studies are context, not a common benchmark or a performance ranking.

Study	Representation/proposal	Evaluation scope
DeepCAD	Learned CAD-operation sequences	Autoencoding and shape generation
Text2CAD	Text-conditioned CAD sequences	Visual, parametric and geometric metrics
ST5 report	Evolved antenna encodings	Simulation, fabrication and prototype comparison
FunSearch	Language-model program proposals	Mathematical/algorithmic evaluators
This study	Fixed CAD schema; retained search	Retrospective rankings and numerical-assessment evidence

B. Evolutionary Engineering and Diversity

Evolutionary engineering predates language-model assistance. Lohn *et al.*'s 2003 ST5 antenna report describes two evolutionary representations, fabricated prototypes and comparison with a hand-designed antenna [4]. Flight qualification was still pending in that report. Its relevance is the connection between representation, computational assessment and physical comparison, rather than evidence that Jarambit has achieved the same experimental maturity.

MAP-Elites organizes retained solutions by behavioral descriptors, preserving alternatives in different niches [5]. Scalar preference and diversity retention answer different questions: which candidate is preferred, and which alternatives remain available? An archive is selected by construction; its distribution is not the distribution of all proposals. Jarambit contains a quality-diversity lane, but the retained cohorts do not establish a common algorithm or descriptor regime across every wave.

C. Agents as Proposal Mechanisms

FunSearch separates language-model program proposals from execution-based evaluation in mathematical and algorithmic settings [6]. Text2CAD demonstrates text-conditioned CAD generation, but does not by that fact establish autonomous mechanical validation. Together these studies motivate a question about proposal interfaces and evaluators, not a demonstrated agent advantage in Jarambit. Table I distinguishes their evaluation scopes.

An engineering agent could assist with schema documentation, source navigation and typed representation proposals. A proposed grammar or decoder revision also changes the evaluator's domain. Such a proposal needs review and independent assessment before its outcomes can be compared with the original representation. Section IX develops this prospective research boundary.

D. Verification, Validation, and Identity

Code verification examines implementation against reference problems; solution verification concerns numerical error for a specified model and observable; validation concerns agreement with the physical system in a declared domain. NASA's models-and-simulations standard supplies credibility context, without a claim of compliance here [7].

The assessment uses scikit-fem for finite-element assembly and a separate DOLFINx path [8], [9]. The DOLFINx methods reference describes the software architecture; the archived records identify the specific 0.9.0 environment. A software citation does not certify this model or establish that its outputs were independently reproduced. Matched solvers can share model and discretization errors, which makes their agreement a different observation from convergence or validation.

Content hashes bind bytes, while model identity, execution records and observable definitions supply their meaning. This distinction is essential when geometry, scorecards and fields originate from different revisions.

E. Additive Manufacturing as Opportunity and Constraint

NIST's additive-manufacturing test artifact addresses geometric characterization through manufactured features and measurements [10]. It supports separating geometric expressiveness, nominal CAD precision and measured process capability. Additive manufacturing may make more complex forms practical to investigate; digital detail alone does not establish as-built accuracy.

This study's contribution lies between these research areas: it reconstructs how retained search rankings relate to subsequent numerical-assessment evidence in one executable CAD system. Learned CAD generation, agent advantage and manufacturing accuracy remain separate empirical questions.

III. From a 129-Column Genome Record to an Identifiable CAD

A. A Gene Is a Contract, Not Merely a Number

A gene requires an explicit coordinate index, physical or discrete type, unit, box domain, coupling law, and decoder revision. A serialized floating-point token alone cannot distinguish a millimeter normal displacement from an angular bevel setting, a discrete feature-count selector, or a binary structural-layer switch. Those semantics determine whether a decoded B-rep or finite-element mesh is reproducible across software revisions.

Jarambit's native crescent representation encodes 128 continuous numerical coordinates paired with one categorical alloy state, and decodes them into a planform and 2.5D thickness field. In every evaluated population table across Cohort 3 (Run 2–Run 11, 1,954,560 evaluations across 10 retained runs of 801 generations) and Cohort 1 (Run 1, 1,984 historical births), each row contains 146 columns: 17 scalar evaluation and search-metadata columns (columns 0..16), followed by 128 numerical `f64` gene coordinates (columns 17..144) and one categorical alloy column (`material`, column 145, taking values `Ti-6Al-4V` or `Ti-3Al-2.5V`), yielding 129 genome-and-material fields per birth record.

B. The Four Structural Blocks of the 128 Numeric Genes

The 128 numeric coordinates partition into four functional blocks that reflect the structural hierarchy of the decoder:

1) **Block A: Silhouette Normal Control Points (`cp00–cp39`, indices** $0..39$**, 40 coordinates).** Each coordinate specifies a normal displacement in millimeters from the 40-station reference outline. Station-specific bounds restrict displacements to $\pm 1.5\,\mathrm{mm}$ on the outer ring arc ($i \in [0,4] \cup [36,39]$), $\pm 2.0\,\mathrm{mm}$ along the tip approach ($i \in [15,22]$), and $\pm 3.5\,\mathrm{mm}$ along the belly, spine, and handle arches.
2) **Block B: Planform, Stock, Bevel, Relief, Lattice, Fuller, and Edge Coordinates (indices** $40..89$**, 50 coordinates).** Indices $40..44$ parameterize global planform and ring dimensions (blade drop angle, grip offset Δx_{grip}, ring wall thickness, ring inner diameter, and neck fillet radius); indices $45..49$ define stock thickness t_{stock} ($[4.5, 6.0]\,\mathrm{mm}$), tip thickness t_{tip} ($[2.2, 5.0]\,\mathrm{mm}$), taper origin $x_{0,\text{taper}}$, and inside/spine bevel angles; indices $50..89$ parameterize the comet groove, ergonomics, three-tier serration depths, handle relief pocket (depth, width, and exponent), dimple lattice, blade fuller, handle flute, surface texture mode, cord wrap, grip pad, and cutting-edge thickness ($[0.25, 1.20]\,\mathrm{mm}$).
3) **Block C: Internal-Machining Coordinates (indices** $90..116$**, 27 coordinates).** These 27 coordinates control triangular isogrid pocketing (depth, pitch, grading, handle start, and land width), pierced skeletonization windows (window count through decay rate), ring lightening flutes (depth, width, and arc span), and angled micro-perforation slots (depth through handle start).
4) **Block D: Continuous-Rail Layer (indices** $117..127$**, 11 coordinates).** The final 11 numeric coordinates encode the binary continuous-rail layer switch $p_{1,\text{on}} \in \{0,1\}$, continuous perimeter rail width w_{rail} ($[2.2, 5.5]\,\mathrm{mm}$), solid neck length l_{neck} ($[6.0, 24.0]\,\mathrm{mm}$), interior pocket depth and corner radius ($d_{\text{pocket}}, r_{\text{pocket}}$), active spline-knot count $k_n \in \{3,4,5\}$, and five interior thickness knots `k0–k4` ($[1.5, 6.0]\,\mathrm{mm}$).

C. Decoder Coupling, Clamping, and Inactive Coordinates

These 129 fields are not independent degrees of freedom. Before phenotype construction, the decoder clamp enforces box bounds on all 128 numeric coordinates together with deterministic cross-gene coupling laws:

$$\begin{aligned} t_{\text{tip}} &\le t_{\text{stock}}, & x_{1,\text{comet}} &\ge x_{0,\text{comet}} + 15\,\mathrm{mm}, \\ s_{1,\text{fuller}} &\ge s_{0,\text{fuller}} + 0.12, & s_{1,\text{win}} &\ge s_{0,\text{win}} + 0.12, \end{aligned} \tag{1}$$

while rounding discrete coordinates (surface texture mode $\in \{0,1,2\}$, window count $\in \{0,\ldots,8\}$, continuous-rail switch $p_{1,\text{on}} \in \{0,1\}$, and spline-knot count $k_n \in \{3,4,5\}$). When $p_{1,\text{on}} < 0.5$, the continuous-rail thickness law is inactive in the decoder even though its 11 numeric coordinates remain serialized in the genome record; conversely, when $p_{1,\text{on}} \ge 0.5$, the continuous perimeter rail and spline-knot thickness law supersede legacy interior pocket features. Preserving all 129 columns losslessly is therefore essential to distinguish an inactive coordinate mutation from a geometric change.

D. Candidate, Genome, and Export Identity

Because multiple birth events can produce identical clamped coordinates (for example, Cohort 1 Run 1 records 1,984 births across 1,965 distinct gene serializations), Jarambit separates

birth/individual identity (Individual ID), genome serialization identity, population-slot appearance, and exported CAD/assessment bundle identity. When comparing parent and child rows across the 128 numeric CSV coordinates (columns 17..144, `cp00`–`k4`), changed-coordinate counts are defined by exact floating-point inequality ($\Delta \neq 0$) on the stored CSV tokens, and raw-coordinate Euclidean L_2 norms across those 128 columns mix heterogeneous units (millimeters, degrees, dimensionless ratios, and discrete selectors) rather than measuring physical geometric distance. For Individual 2000, the STEP solid (1 solid, 1 shell, 55,698 face tokens), the render mesh (44,994 vertices, 90,000 triangles), the $0.25\,\mathrm{mm}^2$ plate mesh (13,503 nodes, 25,209 triangles), and the P_2 tetrahedral solid tip submodel (91,008 tetrahedra, 416,583 displacement degrees of freedom) represent distinct discretizations with separate provenance bindings documented in Section X.

IV. Evolutionary Process, Variation Operators, and Population Evidence

A. Implemented Variation Operators: Block-Preserving Crossover and Six-Mode Mutation

Rather than treating evolutionary search as an opaque black-box optimizer, Jarambit records the exact variation operators applied across its island populations:

- **Two-parent crossover:** To prevent non-physical saw-tooth discontinuities along the closed silhouette spline, the 40 normal control points (`cp00`–`cp39`, indices 0..39) are inherited as a single contiguous block from parent b with probability 0.5 and otherwise retained from parent a. The remaining 88 numeric feature coordinates (indices 40..127) and the categorical alloy grade (`material`) are inherited independently with probability 0.5 per coordinate, followed immediately by deterministic clamping.
- **Six-mode structured mutation:** Each mutation call draws an integer mode op $\in \{0, 1, 2, 3, 4, 5\}$ uniformly at random with step scale $\sigma = 0.15$ (coarse) or $\sigma = 0.04$ (fine):
  1) *Mode 0 (Sparse per-gene Gaussian):* Perturbs each coordinate $i \in [0, 127]$ independently with probability 0.15 by $\mathcal{N}(0, 1) \cdot (\mathrm{hi}_i - \mathrm{lo}_i)\sigma$.
  2) *Mode 1 (Smooth silhouette bump):* Applies a periodic Gaussian normal displacement bump $A \exp(-d^2/(2w^2))$ across all 40 control points around a random station $c \in [0, 39]$ with width $w \in [1.5, 5.5]$.
  3) *Mode 2 (Thickness, taper, and relief operator):* Perturbs stock thickness t_{stock}, tip thickness t_{tip}, taper origin $x_{0,\text{taper}}$, and handle relief depth and width, and—when $p_{1,\text{on}} \geq 0.5$—also perturbs perimeter rail width w_{rail}, pocket depth d_{pocket}, and one interior spline knot `k0`–`k4`.
  4) *Mode 3 (Dimple-lattice operator):* Deactivates the lattice ($d_{\text{lattice}} = 0$) with probability 0.15 or perturbs lattice depth, pitch, fill fraction, angle, and handle/blade extents (68..73).
  5) *Mode 4 (Fuller, flute, texture, and comet operator):* Perturbs blade fuller and handle flute geometry (74..78), surface texture mode and pitch/angle (79..82), and comet groove width/bow (52..54).
  6) *Mode 5 (Internal pocketing or continuous-rail operator):* When $p_{1,\text{on}} \geq 0.5$, mutates the continuous perimeter rail and knot coordinates (118..127) with probability 0.7; otherwise selects one of four sub-operators targeting triangular isogrid (90..98), pierced windows (99..104), ring flute (105..109), or angled micro-perforations (110..116).

  In coarse mode, the alloy grade flips between `Ti-6Al-4V` and `Ti-3Al-2.5V` with probability 0.03 before clamping.

B. Four Distinct Population Collections

Scientifically credible reporting requires distinguishing four collections that share candidates but have different denominators and selection biases:

1) **Birth collection:** Every proposed candidate evaluated or rejected at birth:
   - *Cohort 3 (10-Run 800-Gen Population,* 1,954,560 *Evaluations):* Run 2 through Run 11 each evaluate 195,456 births across generations 0–800 (801 distinct generations per run), totaling 1,954,560 complete 146-column birth records (754,236 search-legal).
   - *Cohort 1 (Historical 17-Gen, Run 1):* 1,984 births across generations 0–16 (1,455 with completed plate finite-element solves, 529 not evaluated by finite elements, and 446 search-legal).
   - *Cohort 2 (Intermediate Screening):* 3,649 multi-stage birth proposals across 24 screening runs, with 2,259 joined to Stage-1 screening observations and 1,390 preserved in an explicit unscreened observation band.
2) **Island-membership (survivor) collection:** Candidates occupying active island slots at generation end, including elite retention and periodic ring migration. In Cohort 1 (Run 1), 2,176 slot occurrences are recorded across Islands 1–4 (4×32 slots over 17 generations), comprising 1,984 initial placements, 192 elite reappearances, and 12 ring-migration events (4 events each at generations 4, 8, and 12).
3) **Quality-diversity archive:** Representatives retained per behavioral niche cell across each run.
4) **Independent Stage-2 / strict assessment collection:** The subset of top-ranked or separately exported candidates admitted to multi-gate CAD, plate, and quadratic solid verification (23 subjects in the historical battery plus the single assessed export Individual 2000 in the automatic certification leaderboard, 1 assessed candidate, 0 qualified, no promoted champion).

C. Recorded Multi-Hop Inheritance and Coordinate-Level Genome Diffs

Figure 2 traces the recorded three-hop ancestry of Individual 1095 (Cohort 1, Run 1, generation 9, highest stored birth fitness fit = 1.15727) directly from the Cohort 1 evaluation ledger and 146-column population table (exact floating-point inequality $\Delta \neq 0$ across the 128 numeric gene columns 17..144;

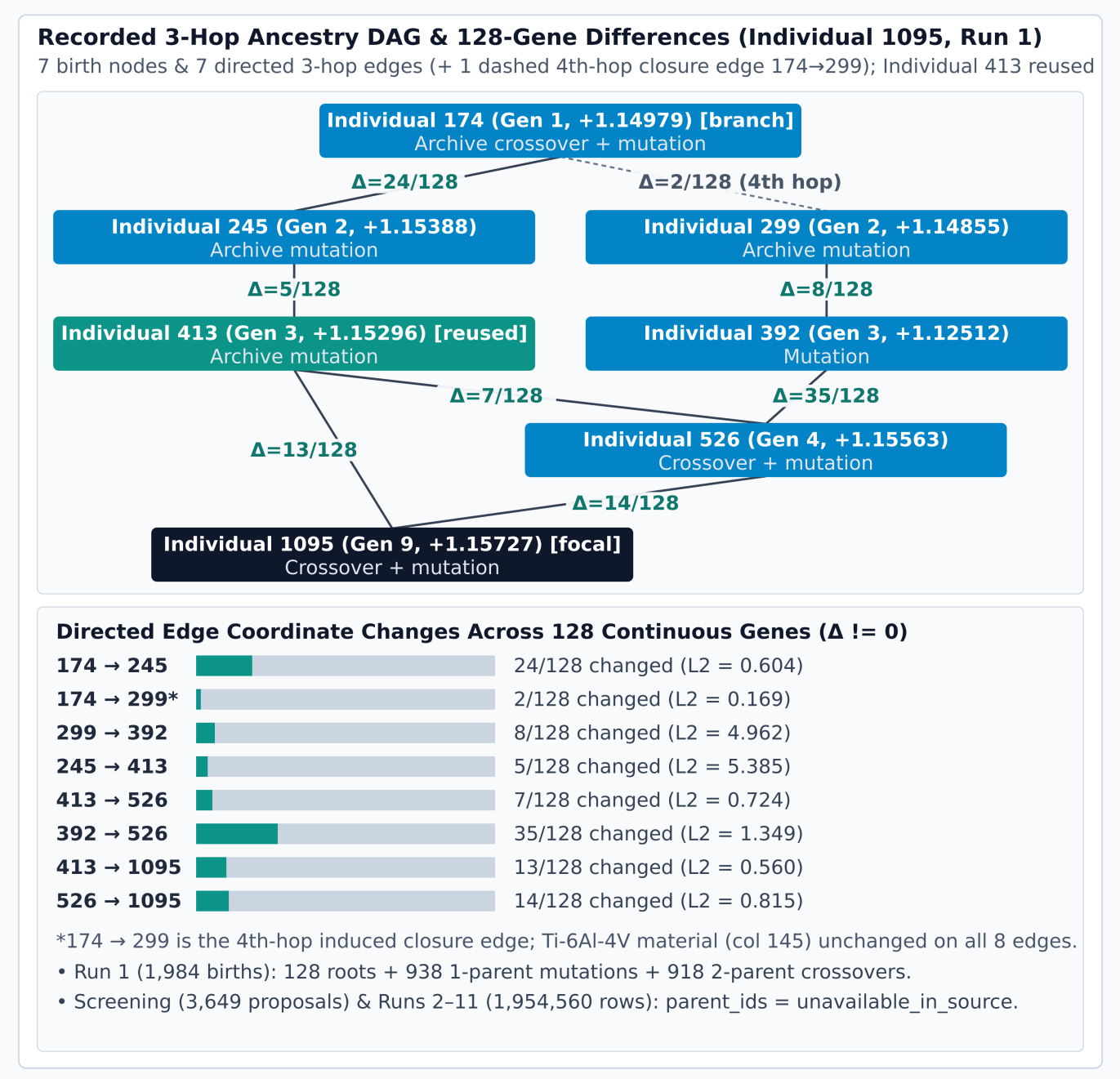


Fig. 2. Cohort 1 (Run 1) recorded three-hop ancestry and 128-coordinate genome differences around Individual 1095 (generation 9, fit = 1.15727, strict verification not met): the 7 retained nodes, 7 three-hop parent edges (solid), and fourth-hop induced edge 174 → 299 (dashed), showing ancestor reuse of Individual 413 (3-hop) and Individual 174 (induced closure) plus exact $\Delta \neq 0$ coordinate counts across CSV columns 17..144.

raw-coordinate Euclidean L_2 norms mix millimeter, degree, and dimensionless units and are not physical geometric distances). Rather than a tree of independent ancestors, the 7 retained individuals form a directed acyclic graph with 7 three-hop parent edges and 1 fourth-hop induced edge (8 induced edges total):

- **Individual 174** (generation 1, fit = 1.14979, archive crossover + mutation from parents 57 and 61) branches in the induced subgraph to produce both **Individual 245** (generation 2, fit = 1.15388, archive mutation from sole parent [174], 24/128 changed numeric coordinates, raw-coordinate $L_2 = 0.604263$, $\max|\Delta| = 0.361541$ across silhouette control points `cp01`–`cp26`) and, via the fourth-hop induced edge 174 → 299, **Individual 299** (generation 2, fit = 1.14855, archive mutation from sole parent [174], 2/128 changed coordinates: lattice depth $\Delta = -0.129784\,\text{mm}$ and lattice pitch $\Delta = +0.108820\,\text{mm}$, raw-coordinate $L_2 = 0.169369$).
- **Individual 245** mutates into **Individual 413** (generation 3, fit = 1.15296, archive mutation, sole recorded parent [245], 5/128 changed coordinates—comet terminal width, cairn depth, handle flute depth, surface texture depth, and

texture angle $\Delta = +5.371762°$—with raw-coordinate $L_2 = 5.385480$), while **Individual 299** mutates into **Individual 392** (generation 3, fit $= 1.12512$, mutation, sole recorded parent $[299]$, $8/128$ changed coordinates, raw-coordinate $L_2 = 4.962176$, $\max|\Delta| = 4.850458$). Individual 413 does *not* descend from Individual 392.

- **Individual 526** (generation 4, fit $= 1.15563$, crossover + mutation, recorded parents $[413, 392]$) differs in $7/128$ numeric coordinates from parent 0 Individual 413 (raw-coordinate $L_2 = 0.724248$, $\max|\Delta| = 0.648528$) and in $35/128$ coordinates from parent 1 Individual 392 (raw-coordinate $L_2 = 1.349493$, $\max|\Delta| = 0.894505$).
- **Individual 1095** (generation 9, fit $= 1.15727$, crossover + mutation, recorded parents $[413, 526]$) reuses ancestor Individual 413 across two distinct paths ($413 \to 1095$ and $413 \to 526 \to 1095$), differing in $13/128$ coordinates from parent 0 Individual 413 (raw-coordinate $L_2 = 0.560427$, $\max|\Delta| = 0.277590$) and in $14/128$ coordinates from parent 1 Individual 526 (raw-coordinate $L_2 = 0.815184$, $\max|\Delta| = 0.648528$; $11/128$ coordinates differ from both parents simultaneously, 3 match only 413, and 2 match only 526, while column 145 remains `Ti-6Al-4V` across all 8 edges).

Crucially, when Individual 1095 was admitted to strict multi-model verification, it failed the regional failure-sequence requirement, demonstrating why high search-stage fitness and recorded lineage do not establish engineering qualification.

V. Fitness Profiles, Priority Groupings, and Stage Semantics

A. Three Distinct Score Regimes and Forbidden Cross-Profile Splicing

Because different search cohorts and assessment stages evaluate different mathematical functions, a common word such as "fitness" does not establish comparability across runs. Jarambit's score registry enforces strict separation across three profile families, prohibiting cross-profile splicing:

1) **Native Quality-Diversity Search Fitness (f, Cohort 1 Run 1 and Cohort 3 Run 2–Run 11):** In the four-island crescent search engine, native fitness f (higher is better) applies conditional legality and feasibility branches: geometrically illegal or unmeshed candidates receive unbounded negative penalty scores ($f < 0$, down to -207.23 in Cohort 1), whereas legal candidates combine normalized mechanical/mass terms with archive-contextual novelty computed against the contemporaneous population and archive. Because novelty is re-evaluated as the archive evolves, f is a contextual search observation rather than an intrinsic physical constant of a genome.
2) **Retained Weighted Reporting Profile ($F \in [0, 100]$, Cohort 2 Reporting Curves):** In the multi-stage ultralight reporting profile, twenty normalized component terms $s_i \in [0, 1]$ are combined with percentage weights summing to 100:

$$F = \sum_{i=1}^{20} w_i s_i, \qquad 0 \le s_i \le 1, \quad \sum_{i=1}^{20} w_i = 100. \tag{2}$$

The twenty terms are organized into four priority groups:

- *Mass and compactness (30%):* ultralight mass and holder/envelope compactness.
- *Form and comfort proxies (25%):* karambit ring/claw archetype, silhouette similarity, and bore-collar geometry proxy.
- *Exported geometry (15%):* watertight single-solid B-rep topology, exported edge sharpness, and point profile.
- *Response and verification (30%):* lateral loads (180 N and 260 N coarse and refined), ring traction (1500 N), body and local tip torsion ($1.82\,\mathrm{N \cdot m}$), quadratic solid tip response and cross-solver agreement, elastic support/buckling, regional yield-order sequence (≥ 1.2), three-level plate mesh study, human-maximum overload, and Goodman fatigue screen.

Within the six retained Cohort 2 reporting curves (15 generation summaries), Individual 2002 records the highest retained weighted reporting profile value ($F = 74.207299$, mass 101.139 g raster / 101.138 g solid). Several terms in F are clipped ratios, Boolean indicators, or screening proxies; a higher value of F orders candidates for research attention without establishing qualification.

3) **Shortfall-Penalized CAD Score (W_{CAD}) and Global Composite (G_1):** In 16-gate CAD evaluation, candidates satisfying all 16 legacy engineering gates receive a positive score in $[1.0, 2.0]$ weighted by mass (0.70), archetype (0.12), survival buffer (0.10), sequence buffer (0.05), and carry stability (0.03):

$$\begin{aligned} W_{\text{CAD}}^{\text{pass}} = 1.0 &+ 0.70\, s_{\text{mass}} + 0.12\, s_{\text{karambit}} \\ &+ 0.10\, s_{\text{survival}} + 0.05\, s_{\text{seq}} + 0.03\, s_{\text{carry}}, \end{aligned} \tag{3}$$

whereas candidates failing any gate receive a negative shortfall penalty:

$$W_{\text{CAD}}^{\text{fail}} = -\left(1.0 + 4.0 \sum_{k \in \text{shortfalls}} \Delta_k\right) + 0.10\, s_{\text{mass}} < 0. \tag{4}$$

Alongside W_{CAD}, the dimensionless composite score G_1 (higher is better, with no physical millimeter unit and no 0.45 cutoff gate) evaluates:

$$\begin{aligned} G_1 = 0.35\, &\min(\text{SF}_{\text{solid_tip}},\, 2.0) + 0.35\, \frac{\text{SPG}}{4.0} \\ &+ 0.20\, s_{\text{aes}} - 0.05\, \frac{\max(0,\, m_g - 77.0)}{10.0} \\ &- 0.05\, \frac{\max(0,\, 1.2 - r_{\text{seq}})}{1.2}. \end{aligned} \tag{5}$$

For the separately assessed controlled export Individual 2001, $W_{\text{CAD}} = -3.878704$, $G_1 = 0.623903$, and mass $= 75.382$ g (75.1992 g 3D solid), meeting 11 of 16 legacy gates and failing 5; for companion controlled export Individual 2007, $W_{\text{CAD}} = -3.964924$, $G_1 = 0.621634$, and mass $= 75.642$ g (75.5049 g 3D solid), likewise meeting 11 of 16 legacy gates and failing the same 5 gates while failing grouped verification.

TABLE II
COHORT 3 TEN-RUN 800-GENERATION CENSUS (RUN 2–RUN 11; 195,456 EVALUATED BIRTHS AND 3,204 SUMMARY ROWS PER RUN; 1,954,560 TOTAL)

Run	Legal Births (%)	Max f	Min $L = -f$
Run 2	74,874 (38.31%)	1.1327	−1.1327
Run 3	77,011 (39.40%)	1.1089	−1.1089
Run 4	77,396 (39.60%)	1.1380	−1.1380
Run 5	77,625 (39.71%)	1.1440	−1.1440
Run 6	70,278 (35.96%)	1.1156	−1.1156
Run 7	74,777 (38.26%)	1.1228	−1.1228
Run 8	71,550 (36.61%)	1.1108	−1.1108
Run 9	76,284 (39.03%)	1.1408	−1.1408
Run 10	74,309 (38.02%)	1.1186	−1.1186
Run 11	80,132 (41.00%)	1.1177	−1.1177
Total / Range	754,236 (38.59%)	1.1089–1.1440	−1.1440..−1.1089
Mean $\pm s$	$75{,}423.6 \pm 2{,}936.6$	1.1250 ± 0.0129	-1.1250 ± 0.0129

B. Reversible Display-Loss Conventions

To support minimization-oriented progress plots without altering underlying stored evidence, each profile defines an explicit order-preserving display loss L:

$$L_{\text{native}} = -f, \qquad L_{\text{ultralight}} = 100 - F, \qquad L_{\text{CAD}} = -W_{\text{CAD}}. \tag{6}$$

For cumulative best-so-far series $H_g = \max_{0 \le k \le g} r_k$, the corresponding display-loss envelope $-H_g$ or $100 - H_g$ is monotone non-increasing by construction. All figures declare both the display-loss transform and the native score values, preserving negative penalty tails without clipping.

C. Cohort 3 Cross-Run Trajectory Comparison (Run 2–Run 11)

Table II summarizes the ten retained 800-generation four-island runs in Cohort 3 (1,954,560 total evaluated births and 32,040 generation-by-island summary rows). Every run evaluated 195,456 distinct candidate IDs across 801 generations (0–800). Search-legal births range from 70,278 (35.96%, Run 6) to 80,132 (41.00%, Run 11), totaling 754,236 (38.59%; mean $75{,}423.6 \pm 2{,}936.55$ sample SD, population SD 2,785.86) across the ten-run ensemble, while run-maximum native fitness reaches 1.1089 (Run 3) to 1.1440 (Run 5) (mean 1.1250 ± 0.01285 sample SD, population SD 0.01220).

D. Retained Spatial Plate Fields and Saliency Crops for Individual 2000

The retained $0.25\,\mathrm{mm}^2$ plate artifact for Individual 2000 (13,503 nodes, 25,209 triangles) stores seven per-triangle arrays: five structural demand maps—the scaled multi-case von Mises envelope (MPa, maximum $1{,}388.6695\,\mathrm{MPa}$, with 75 cells inside the $4\,\mathrm{mm}$ tip exclusion forced to zero by the producer and rendered in neutral gray), unit out-of-plane lateral bending response ($\mathrm{MPa/N}$, 25,209 active cells), unit in-plane tip-pry T6 membrane response ($\mathrm{MPa/N}$, 25,209 active cells; in the multi-case envelope construction, both lateral and pry unit responses are multiplied by $260\,\mathrm{N}$), unit ring-traction T6 centroid response ($\mathrm{MPa/N}$, 10,334 active cells and 14,875 seated-region masked cells, scaled by $1500\,\mathrm{N}$ in the envelope), and unit chord-torsion response ($\mathrm{MPa}/(\mathrm{N\,mm})$, 25,209 active cells, scaled by $1820\,\mathrm{N\,mm}$ in the envelope)—together with two auxiliary fields: per-element safety factor (dimensionless) and local element thickness (mm). Figure 3 displays the retained demand maps, while Figure 4 pairs the $\geq 90\%$ integrated-saliency local crops with whole-object picture-in-picture locators (and the unlit 2D 5-map percentile composite reference plate with contributor support $q(x) \in \{3/5, 4/5, 5/5\}$ is archived in the linked Technical Supplement, Section X).

VI. NUMERICAL OBSERVATION FAMILIES AND METHODOLOGICAL EVOLUTION

EVALUATING an evolved physical geometry requires translating continuous shape variations into concrete geometric and mechanical observables. Earlier internal documentation listed forty-four catalog predicates—sixteen legacy diagnostic checks, ten grouped requirement predicates, and eighteen historical battery columns—inline alongside a 414-cell categorical status matrix. While complete, that administrative taxonomy obscured the physical and numerical methods.

In this section, we organize the main-text presentation around the **eleven substantive observation and model families** defined by the underlying physics and software architecture (Table III). Every subsection follows a uniform seven-part structure: (1) engineering question, (2) represented observable and units, (3) numerical domain and formulation, (4) role across historical and weighted objective regimes, (5) retained visual evidence for the assessed export Individual 2000, (6) interpretation limits, and (7) cross-reference to the Supplement technical crosswalk (Section X). The complete 23-subject by 18-evaluator historical battery matrix (221 stored passes, 81 failures, and 112 blocked cells; preserved in the linked Supplement generated/battery-matrix.svg and Section X) and the item-by-item 44-entry crosswalk are preserved without omission.

A. Gate and Objective Evolution Across Regimes

Understanding the retained data requires distinguishing three evaluation regimes that coexist in the repository without conflating their semantics:

1) **Historical Native Search (Cohort 1 Run 1, Cohort 3 Run 2–Run 11, and Four-Island Quality-Diversity):** During native quality-diversity evolution, each birth is decoded and checked sequentially for geometric legality, reference-contour similarity, and plate finite-element feasibility. The historical scalar fitness f is therefore *piecewise and conditional*: candidates failing geometric legality or the reference-contour floor receive negative penalty branches without invoking FEM (529 of 1984 births in Cohort 1 Run 1), whereas evaluated candidates combine normalized structural response ratios, mass terms, and archive-contextual novelty. Removing the categorical pass/fail matrix from the opening narrative does not make the historical search algorithm gate-free; its conditional branch structure is preserved as an explicit methodological fact.
2) **Multi-Stage Screening and Retained Weighted Reporting Profile:** In the intermediate pipeline, Stage 1



Fig. 3. Individual 2000 retained $0.25\,\mathrm{mm}^2$ plate arrays (25,209 triangles, 13,503 nodes; single assessed export, 1 assessed, 0 qualified). A: multi-case stress envelope (MPa). B: unit lateral bending response (MPa/N). C: unit ring-traction response (MPa/N). D: unit torsion response (MPa/(N mm)). Gray cells denote producer-masked exclusions (75 tip cells in A; 14,875 seated-ring cells in C), not zero physical stress.

screens candidates using fast plate and geometric observables, while subsequent evaluation adds local quadratic-solid and refined observations. The 20-term retained weighted reporting profile $F = \sum_{i=1}^{20} w_i s_i \in [0, 100]$ aggregates four thematic groups (30% mass/compactness, 25% form/comfort proxies, 15% exported geometry, 30% response/verification). Prior to repository revision `72e25a360470`, selection and reporting differed on two placeholder geometry inputs, and round maxima combined stage values across rounds. We therefore keep historical native f, Stage-1 screen scores, and the retained weighted reporting profile strictly separate.

3) **Fixed Historical Assessment Battery (23 Subjects × 18 Columns):** The historical CAD battery evaluates 23 fixed subjects (21 earlier-cohort subjects and two controlled-probe exports including Individual 2001) across 18 heterogeneous columns mixing shared software controls (92 of the 221 passes), missing geometry-binding receipts (112 blocked cells across unit-declaration, single-solid, and pin-clearance checks plus four reference rows), record-completeness flags (20 quadratic-solid file-completeness records), and strict grouped requirement gates (19 failures and 4 blocked reference rows).

B. *Family 1: Representation and Exported Geometry*

Question & Observables: Does a decoded genome produce a closed, single-solid B-rep and surface mesh with verifiable unit scaling, mass (g), edge/tip profile geometry (mm), and cryptographic byte binding? **Domain & Formulation:** The decoder maps the 128-gene vector and material density into a 2D mid-surface contour, variable thickness/bevel fields, and 3D STEP/STL exports. **Objective Role:** Geometric legality gates initial evaluation in native QD search; exported geometry terms contribute 15% of the retained weighted reporting profile. **Retained Visual & Limits:** The linked Supplement (generated/export-identity.svg, Section X) and Figure 1 bind

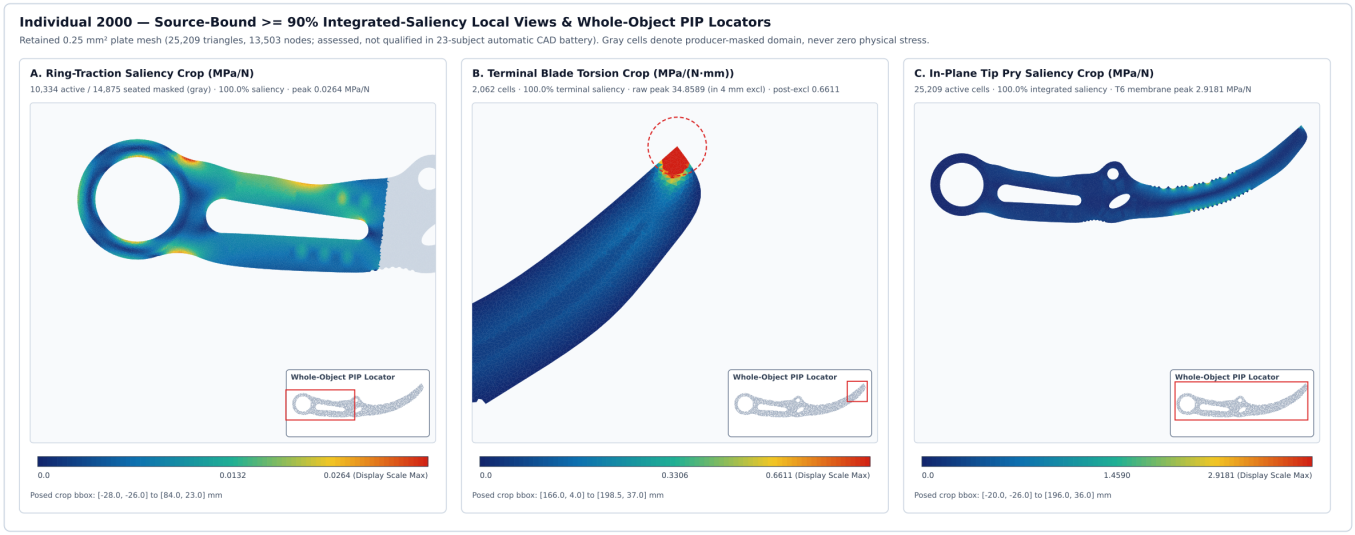


Fig. 4. Individual 2000 $\geq 90\%$ area-weighted integrated-saliency local crops with whole-object picture-in-picture locators (25,209 triangles, 13,503 nodes; Panels A–C): ring traction ($\mathrm{MPa/N}$), terminal blade-tip torsion ($\mathrm{MPa/(N\,mm)}$), and multi-case von Mises envelope (MPa).

Individual 2000's genome, STEP/STL digests, 1 solid / 1 shell / 55,698 STEP face tokens, OpenCascade valid B-rep record, and 44,994-vertex / 90,000-triangle render mesh. Three geometry checks in this family (holder-envelope clearance, ring/claw planform proportions, and B-rep edge/tip metrology) intentionally omit synthetic subsection visuals (4 entries total across the 44-entry catalog) because nominal parameters and exported B-rep metrology must not be conflated. In the historical battery, unit-declaration, single-solid, and pin-clearance bindings are blocked across all 23 subjects (Supplement Section X).

C. Family 2: Shape and Reference Comparison

Question & Observables: How closely does the decoded exterior silhouette match the authored reference archetype in Hausdorff distance (mm), curvature-histogram Earth Mover's Distance (EMD, dimensionless), and composite similarity score $\in [0, 1]$? **Domain & Formulation:** Evaluates the closed 2D exterior polyline (4,230 points for Individual 2000) against the retained C.5 exterior reference polyline (1,117 points, reused by C.6) under a common isotropic coordinate transform. **Objective Role:** Enforces the reference-contour similarity floor before FEM in native QD search and supplies form/reference proxy terms in weighted reporting. **Retained Visual & Limits:** The linked Supplement (generated/reference.svg, Section X) displays both polylines and the saved descriptors (Hausdorff $2.285\,\mathrm{mm}$, curvature EMD 0.0071, similarity score 0.4079, empty failure list). Reference resemblance measures conformity to an authored geometric prior, not human ergonomic approval or comfort.

D. Family 3: Plate Lateral and Pry Response

Question & Observables: What are the linear-elastic unit responses under out-of-plane lateral tip load ($\mathrm{MPa/N}$) and in-plane tip-pry load ($\mathrm{MPa/N}$), and the resulting lateral load capacity F_{lat} (N) across coarse, certification, and strict plate discretizations? **Domain & Formulation:** Discrete Kirchhoff Triangle (DKT) out-of-plane bending elements and six-node quadratic membrane (T6) in-plane bending elements on the 2D mid-surface (13,503 nodes and 25,209 triangles at the $0.25\,\mathrm{mm}^2$ level; minima taken over 0.25 and $0.125\,\mathrm{mm}^2$ for certification, and $0.0625\,\mathrm{mm}^2$ for strict refinement). In the multi-case envelope construction, both lateral bending and tip-pry unit-response fields are multiplied by $260\,\mathrm{N}$. **Objective Role:** Primary structural capacity term in native QD fitness and the response/verification group (30%) of the retained weighted reporting profile. **Retained Visual & Limits:** Figure 3B shows the $0.25\,\mathrm{mm}^2$ lateral unit-response field (25,209 valid triangles), while Fig. 3A shows the scaled multi-case envelope (25,134 valid triangles, with 75 tip-exclusion cells masked in gray; peak $1{,}388.6695\,\mathrm{MPa}$). Figure 5C records the certification ($168.9008\,\mathrm{N}$) and strict ($168.1786\,\mathrm{N}$) lateral capacities against the $180\,\mathrm{N}$ coarse and $260\,\mathrm{N}$ certification/strict thresholds, and no full spatial field is archived for the $0.0625\,\mathrm{mm}^2$ strict mesh (Supplement Section X).

E. Family 4: Ring Traction

Question & Observables: What is the in-plane membrane von Mises stress per unit ring-pull resultant ($\mathrm{MPa/N}$) and the static ring-pull capacity (N)? **Domain & Formulation:** Six-node quadratic membrane triangles (T6) on the mid-surface under static ring-bore traction. In the retained 25,209-triangle plate mesh, 14,875 cells in the seated handle/blade region are producer-zeroed, leaving 10,334 active triangles in the ring and immediate transition domain. **Objective Role:** Contributes to the joint structural floor in native search and multi-stage response scoring. **Retained Visual & Limits:** Figure 3C renders the 10,334 valid triangles and explicitly masks the 14,875 producer-zeroed seated cells in neutral gray so excluded domain is never misread as zero stress. Static elastic ring traction is not a dynamic drop test or human-finger injury model (Supplement Section X).

TABLE III
THE ELEVEN MAIN-TEXT OBSERVATION AND MODEL FAMILIES, THEIR REPRESENTED OBSERVABLES, RETAINED VISUAL BINDINGS FOR INDIVIDUAL 2000, AND SCIENTIFIC VERIFICATION CRITERIA.

Methodological Family	Observable & Units	Domain & Formulation	Retained Visual / Scope	Verification Criteria
1. Representation & exported geometry	B-rep solid/shell/face counts; mesh closure; mass (g); SHA-256 identity	128-gene + material decoder to STEP, STL, and 2D/3D FEM meshes	Fig. 1; Supplement export-identity.svg; 3 nominal/B-rep checks omit synthetic visuals	Watertight single solid; holder envelope; ring/claw & bevel geometry
2. Shape & reference comparison	Hausdorff distance d_H (mm), curvature EMD, similarity score $\in [0,1]$	Closed 2D exterior contour (4,230 pts) vs. C.5/C.6 reference (1,117 pts)	Supplement reference.svg (2.285 mm, EMD 0.0071, score 0.4079)	Reference-contour similarity ($d_H \leq 15\,\mathrm{mm}$, $\mathrm{EMD} \leq 0.12$, score ≥ 0.25)
3. Plate lateral & pry response	Unit bending/pry σ_{vM} (MPa/N); lateral capacity F_{lat} (N)	DKT bending & T6 membrane; both scaled by 260 N in multi-case envelope	Fig. 3 (A, B); Fig. 5 (C: 168.90 vs. 168.18 N)	Lateral bending & pry capacity (180 N coarse; 260 N cert. & refined plate)
4. Ring traction	Unit in-plane σ_{vM} (MPa/N); ring pull capacity F_{ring} (N)	T6 membrane triangles (25,209 total; 10,334 unmasked ring support)	Fig. 3 (C: 14,875 seated cells masked in gray)	Retention-ring pull capacity (1500 N membrane traction & human overload)
5. Body & local torsion	Unit torsional σ_{vM} ($\mathrm{MPa}/(\mathrm{N\,mm})$); body/tip safety factors	DKT plate torsion (1.82 N m scale); 4 mm tip capacity exclusion	Fig. 3 (D); Fig. 5 (C: 0.5239 cert / 0.5287 strict)	Body & tip torsional response (1.82 N m cert. and refined plate torsion)
6. Quadratic solid submodel	Nodal-max & regional safety factor $\mathrm{SF}_{\mathrm{solid}}$ (dimensionless)	3D quadratic tetrahedra (P_2) on terminal submodel; 1.82 N m torsion	Fig. 6 (C–D: scalar pairs & binned profiles; no 3D field)	Quadratic tetrahedral tip submodel (1.82 N m, 10-node P_2 continuum)
7. Support & linear stability	Supported ratio (5.2027) & linear buckling eigenvalue λ_1 (7.0351)	DKT/CST assembly with bore springs & Winkler grip pad; $\mathbf{K}_s^{-1}\mathbf{G}$	Fig. 5 (A: schematic & scalars; eigen-operator audit open)	Compliant bore/grip support & linear buckling ($\mathbf{K}_s^{-1}\mathbf{G}$)
8. Regional yield-order proxy	Elastic sequence ratio r_{seq} (dimensionless, ≥ 1.2 target)	Ratio of governing regional elastic capacities across plate mesh sequence	Fig. 6 (A: 3-level trace; GCI estimate refused)	Tip-first yield-order ratio ($r_{\mathrm{seq}} \geq 1.2$)
9. Fatigue model	Modified-Goodman factor n_f (dimensionless; saved 0.4718)	$n_f^{-1} = \sigma_a/S_e + \sigma_m/S_u$ under $R = 0$ zero-to-max cycle	Fig. 5 (B: scalar only; no spatial fatigue field or life test)	Modified-Goodman cyclic fatigue factor ($R = 0$)
10. Discretization & solver consistency	3-level GCI admission status; matched-mesh solver ΔSF ($< 0.05\%$)	Richardson extrapolation refusal rules; native Rust vs. DOLFINx P_2 solid	Fig. 6 (A–C: metadata valid, GCI refused, $< 0.05\%$ solid match)	Three-level plate mesh convergence & independent-solver P_2 parity
11. Software & completeness controls	Continuous W_{CAD} subscores/shortfalls; contract pass/fail/blocked counts	Analytic invariance, GCI arithmetic fixtures, claim gates, side-study refs	Fig. 5 (D); Supplement battery-matrix.svg	Continuous CAD subscores, shortfall penalties & fail-closed software controls

F. Family 5: Body and Local Plate Torsion

Question & Observables: How do the body region and terminal tip region respond to a $1.82\,\mathrm{N\,m}$ torsional load in unit response ($\mathrm{MPa}/(\mathrm{N\,mm})$) and regional safety factors (SF, dimensionless)? **Domain & Formulation:** DKT plate torsion model with distinct regional masks: body torsion excludes a $4\,\mathrm{mm}$ terminal tip zone (75 cells at $0.25\,\mathrm{mm}^2$) when computing body capacity and the global envelope, whereas local tip torsion evaluates the terminal zone at certification ($0.25/0.125\,\mathrm{mm}^2$) and strict ($0.0625\,\mathrm{mm}^2$) levels. **Objective Role:** Separates global blade/handle torsional stiffness from localized tip fragility across both search and assessment stages. **Retained Visual & Limits:** Figure 3D shows the unmasked raw torsion unit response across all 25,209 triangles with the $4\,\mathrm{mm}$ capacity-exclusion boundary dashed; Fig. 5C pairs the certification (0.5239) and strict (0.5287) local tip safety factors. Two-dimensional Kirchhoff plate theory cannot capture three-dimensional through-thickness shear near a sharp point, motivating Family 6 (Supplement Section X).

G. Family 6: Quadratic Solid Submodel

Question & Observables: What is the three-dimensional continuum safety factor ($\mathrm{SF}_{\mathrm{solid}}$, dimensionless) in the terminal tip region under $1.82\,\mathrm{N\,m}$ torsion? **Domain & Formulation:** Ten-node quadratic tetrahedral (P_2) continuum submodel of the terminal tip region, recording nodal-maximum and regionally binned safety factors. **Objective Role:** High-fidelity tip verification anchor. **Retained Visual & Limits:** Figure 6C–D plots the matched-mesh scalar safety factors and regional longitudinal profiles. No full 3D volumetric stress field array is retained in the publication inputs. Furthermore, in the historical 23-subject battery, the quadratic-solid tip column records 20 passes that signify *record completeness*, not 20 favorable strength margins (Supplement Section X).

H. Family 7: Compliant Support and Linear Stability

Question & Observables: Under compliant bore-bearing springs and a Winkler grip-pad foundation, what are the lateral regional stress ratio and the lowest linear buckling eigenvalue λ_1

(dimensionless)? **Domain & Formulation:** Alternate DKT/CST assembly solving static compliant support (reported ratio 5.2027 vs. multi-case certification comparator 1.4353) and linear eigenbuckling ($\lambda_1 = 7.0351$) via Krylov iteration on $\mathbf{K}_s^{-1}\mathbf{G}$. **Objective Role:** Side-report stability diagnostic in assessment and the historical battery. **Retained Visual & Limits:** Figure 5A presents the support schematic and archived scalars. Because standard Hermitian Lanczos iteration requires an operator self-adjoint in the inner product and symmetric $\mathbf{K}_s, \mathbf{G}$ do not make $\mathbf{K}_s^{-1}\mathbf{G}$ Euclidean-symmetric without a $\mathbf{K}_s$-inner-product formulation or retained eigenpair residual, the numerical-operator audit remains explicitly open (Supplement Section X).

I. Family 8: Regional Yield-Order Proxy

Question & Observables: Does the modeled elastic response ratio between designated body/ring and tip regions (r_{seq}, dimensionless) exceed the 1.2 ordering threshold? **Domain & Formulation:** Quotient of regional linear-elastic capacity factors evaluated across the plate mesh sequence. **Objective Role:** Penalizes tip-first elastic overload in native search and weighted reporting. **Retained Visual & Limits:** Figure 6A plots the sequence proxy across the three plate mesh levels. Because linear elasticity models neither plastic redistribution, crack initiation, nor dynamic fracture propagation, a favorable sequence ratio is an elastic design proxy rather than a proven physical failure sequence (Supplement Section X).

J. Family 9: Fatigue Model

Question & Observables: What is the modified-Goodman endurance safety factor n_f (dimensionless) under a repeated zero-to-maximum load cycle? **Domain & Formulation:** Evaluates $n_f^{-1} = \sigma_a/S_e + \sigma_m/S_u$ with $\sigma_a = \sigma_m = \sigma_{\max}/2$ and Marin-corrected Ti-6Al-4V endurance limit S_e. **Objective Role:** Durability screen (saved scalar $n_f = 0.4718$ for Individual 2000; 19 unfavorable and 4 blocked in the historical battery). **Retained Visual & Limits:** Figure 5B displays the saved scalar 0.4718. No spatial fatigue field is stored, and nominal Goodman factors do not substitute for surface-roughness metrology, residual-stress measurement, or cyclic coupon testing (Supplement Section X).

K. Family 10: Discretization and Separate-Solver Consistency

Question & Observables: Do three-level plate mesh sequences admit an asymptotic Grid Convergence Index (GCI), and do independent finite-element implementations agree on the matched P_2 solid mesh? **Domain & Formulation:** Three-level Richardson extrapolation and refusal predicates plus matched-mesh P_2 tetrahedral solves comparing the native solver against FEniCSx/DOLFINx. **Objective Role:** Verification and credibility gates (Section VII). **Retained Visual & Limits:** Figure 6A–B shows that while mesh-sequence metadata are valid, oscillatory and nonpositive observed orders cause the GCI error estimates to be refused. Figure 6C shows that the two solvers on the identical P_2 mesh agree within 0.05%, confirming solver-implementation parity on a shared discretization without bounding discretization or modeling error (Supplement Section X).

L. Family 11: Continuous Objective Subscores and Result-Completeness Controls

Question & Observables: How do continuous CAD quality subscores and shortfall magnitudes combine into W_{CAD}, and do unit/invariance tests, GCI refusal arithmetic, claim-admission gates, and historical side-report references behave as specified? **Domain & Formulation:** Continuous subscore and shortfall evaluation plus automated positive and negative software fixtures across the historical battery columns. **Objective Role:** Separates continuous design quality from gate shortfalls and fail-closed repository governance; software-control passes are never weighted as mechanical fitness. **Retained Visual & Limits:** Figure 5D presents the five normalized CAD quality subscores ($s_{\mathrm{mass}} = 0.4870$, $s_{\mathrm{karambit}} = 0.4009$, $s_{\mathrm{survival}} = 0.0$, $s_{\mathrm{seq}} = 0.9996$, $s_{\mathrm{carry}} = 0.6000$; pass-branch unpenalized quality score 1.4570) alongside the three continuous gate shortfalls (0.4710 refined tip torsion, 0.3531 refined plate bending, and 0.3307 quadratic solid tip; conditional failed-branch formula $W_{\mathrm{CAD}} = -1.0 - 4.6191 + 0.10 s_{\mathrm{mass}} = -5.5704$; 10 of 16 legacy gates and 2 of 10 requirement groups met) for Individual 2000, while the linked Supplement (generated/battery-matrix.svg and generated/battery-matrix.json, Section X) retains all 414 historical cells across the 23 battery subjects (221 stored passes, 81 failures, and 112 blocked records), explaining why 92 repeated software-control passes across four build columns and 57 shared side-study references must be separated from candidate-specific mechanical tests.

VII. Numerical Verification and Scientific Assurance

A. The Multi-Model Finite-Element Hierarchy

The platform's mechanical evaluation couples a 2.5D plate-and-membrane finite-element formulation with a localized 3D quadratic-solid submodel. At the algebraic level, each linear-elastic static solve evaluates

$$\mathbf{K}\mathbf{u} = \mathbf{f}, \tag{7}$$

where the stiffness matrix $\mathbf{K}$, displacement vector $\mathbf{u}$, and load vector $\mathbf{f}$ depend on the decoded planform, spatially varying thickness field, alloy constants, support idealization, and element formulation. In the plate path, out-of-plane lateral bending ($\mathrm{MPa/N}$) and chord torsion ($\mathrm{MPa/(N\,mm)}$) assemble discrete Kirchhoff triangle (DKT) bending operators paired with constant-strain triangle (CST) membrane terms, whereas the in-plane tip-pry solve ($\mathrm{MPa/N}$) and assessed ring-traction solve ($\mathrm{MPa/N}$) instantiate six-node quadratic (T6) membrane triangles. In the multi-case von Mises envelope construction, both lateral bending and tip-pry unit responses are multiplied by $260\,\mathrm{N}$, alongside $1500\,\mathrm{N}$ ring traction and $1820\,\mathrm{N\,mm}$ torsion. Consequently, the retained ring-response and pry maps for Individual 2000 record T6 centroid von Mises stress per unit load ($\mathrm{MPa/N}$), whereas the lateral and torsion maps record per-triangle maxima over three corner-extrapolated Gauss-point bending stresses.

In the local 3D solid path, the terminal blade region is discretized with ten-node quadratic (P_2) tetrahedra to resolve three-dimensional tip-torsion gradients that a 2.5D plate assumption cannot capture. A richer element formulation



Fig. 5. Individual 2000 saved scalar observations, continuous CAD objective subscores, and shortfall magnitudes (single assessed export in the automatic certification leaderboard, 1 assessed, 0 qualified, no promoted champion). A: compliant support-model schematic and recorded scalars; the eigen-operator audit remains open. B: saved Goodman fatigue factor (0.4718) and human-overload response factors. C: certification versus strict scalar observations (168.9008 N vs. 168.1786 N lateral capacity; 0.5239 vs. 0.5287 local tip torsion). D: continuous CAD quality subscores ($s_{\mathrm{mass}} = 0.4870$, $s_{\mathrm{karambit}} = 0.4009$, $s_{\mathrm{survival}} = 0.0$, $s_{\mathrm{seq}} = 0.9996$, $s_{\mathrm{carry}} = 0.6000$; pass-branch unpenalized quality score 1.4570) paired with continuous shortfall magnitudes (0.4710 refined tip torsion, 0.3531 refined plate bending, 0.3307 quadratic solid tip; failed-branch formula $W_{\mathrm{CAD}} = -1.0 - 4.6191 + 0.10 s_{\mathrm{mass}} = -5.5704$; 10 of 16 legacy gates and 2 of 10 requirement groups met).

resolves local triaxial stress effects, but remains conditioned on the transferred submodel boundary displacements and mesh density.

B. Observable Semantics and Domain Masks

Comparing scalar safety factors or stress fields across solvers requires binding the exact postprocessing observable and regional mask to the mesh. For Individual 2000 (13,503 plate nodes and 25,209 triangles at the $0.25\,\mathrm{mm}^2$ discretization level across seven retained mesh arrays):

- The multi-case von Mises envelope zeroes 75 triangles whose centroids lie within $4.0\,\mathrm{mm}$ of the boundary tip node, leaving 25,134 active triangles with a maximum recorded elastic envelope stress of $1{,}388.6695\,\mathrm{MPa}$.

- The ring-traction field zeroes $14{,}875$ triangles in the seated ring support zone ($x \geq 76 + \Delta x_{\text{grip}} - 3\,\text{mm}$, with one boundary-adjacent triangle retained because its positive stored coefficient proves it was not source-zeroed), leaving $10{,}334$ active triangles.
- The unit-response lateral (MPa/N), tip-pry (MPa/N), and torsion ($\text{MPa}/(\text{N}\,\text{mm})$) fields retain all $25{,}209$ triangles while outlining the $4.0\,\text{mm}$ tip capacity-exclusion boundary; the auxiliary safety-factor and local thickness fields are likewise defined on all $25{,}209$ triangles.

In the dimensionless 5-map percentile composite index (combining the multi-case envelope and unit lateral, pry, ring-traction, and torsion maps), all five maps are simultaneously active on $10{,}334$ triangles ($q(x) = 5/5$), four maps on $14{,}800$ seated-ring triangles ($q(x) = 4/5$), and three maps on 75 tip-exclusion triangles ($q(x) = 3/5$). Every composite field visualization therefore publishes the per-triangle contributor count $q(x)$ alongside the area-weighted midrank percentile value.

C. Three-Level Plate Mesh Study and Refused Grid-Convergence Estimates

A grid-convergence index (GCI) estimate [12] requires asymptotic monotone convergence across a consistent mesh sequence. For Individual 2000, the retained three-level plate study (Figure 6A–B) evaluates maximum triangle areas $A_{\max} \in \{0.25, 1.0, 4.0\}\,\text{mm}^2$ (effective grid spacings $h = \sqrt{A_{\max}} \in \{0.5, 1.0, 2.0\}\,\text{mm}$, with $25{,}209$, $6{,}803$, and $3{,}166$ elements, respectively). Although the mesh-sequence metadata are valid, neither GCI uncertainty estimate is admitted:

- For the regional yield-order sequence proxy (Figure 6A), the three-level series is monotone but yields a negative observed order ($p = -2.9003$), violating asymptotic convergence.
- For the lateral safety-factor series (Figure 6B), the three-level differences oscillate in sign across refinements.

Both Richardson-extrapolated continuum estimates and fine-grid GCI percentages are therefore recorded as null (unadmitted). Separate certification versus strict refinement pairs ($168.9008\,\text{N}$ vs. $168.1786\,\text{N}$ lateral capacity at $0.0625\,\text{mm}^2$; 0.5239 vs. 0.5287 local plate torsion factor) are two-level comparisons and are never presented as three-level GCI studies.

D. Independent-Solver Quadratic Solid Verification

To verify implementation consistency of the local 3D solid path on Individual 2000, two independent finite-element packages (scikit-fem [8] and DOLFINx 0.9.0 [9]) solve the identical saved quadratic tetrahedral mesh ($91{,}008$ ten-node tetrahedra, $416{,}583$ displacement degrees of freedom; Figure 6C). Within the evaluation region ($4 \leq d_{\text{tip}} < 15\,\text{mm}$ and cut distance $> 3\,\text{mm}$):

- Under incident-cell vertex-maximum recovery, the two solvers report safety factors of 0.669 and 0.6693 (relative difference 0.0448%).
- Under incident-cell unweighted vertex-mean recovery, the two solvers report safety factors of 0.726 and 0.7263 (relative difference 0.0413%).

The primary solve converges in 152 iterations (relative residual 8.80×10^{-11}) and the independent sidecar in 158 iterations. This narrow cross-solver agreement verifies algebraic assembly and recovery consistency on the shared mesh, while leaving boundary-transfer idealizations and physical fracture unvalidated.

E. Support Eigenvalue Audit and Paired Assurance Controls

In the elastic support and buckling side study for Individual 2000, adding bore-bearing springs and a Winkler grip-pad foundation yields a lateral regional ratio of 5.2027 and an elastic buckling eigenvalue of 7.0351 (compared with the multi-case certification minimum of 1.4353). Because the producer invoked a symmetric Krylov eigensolver on the product operator $\mathbf{K}_s^{-1}\mathbf{G}$ without retaining an explicit Euclidean operator-symmetry check or eigenpair residual, that numerical-verification question remains explicitly flagged as open. More broadly, the repository's frozen scientific-assurance catalog specifies 178 challenge definitions (356 paired positive and negative control slots across 20 families) to ensure that corrupted hashes, non-convergent mesh sequences, and synthetic fixtures fail closed rather than silently passing.

VIII. Retrospective Results Across Three Cohorts

A. Cohort 3: Ten-Run 800-Generation Population Census (Run 2–Run 11)

Figure 7 summarizes the ten retained four-island quality-diversity runs that completed generation 800 (Run 2 through Run 11) alongside the separate historical (Run 1) and intermediate screening regimes ($1{,}954{,}560$ native + $1{,}984$ historical + $2{,}259$ screened $= 1{,}958{,}803$ numeric scored observations plus $1{,}390$ unscreened proposals in a separate band $= 1{,}960{,}193$ admitted records across separate search regimes, 0 excluded). Each Cohort 3 run evaluated $195{,}456$ births across 801 generations (0–800) and $3{,}204$ generation-by-island summary rows (4 islands $\times 801$ generations), yielding $1{,}954{,}560$ complete 146-column birth records and $32{,}040$ summary rows in total. Across the ten retained runs, $754{,}236$ evaluations (38.59%) satisfy search-level geometric legality ($610{,}051$ with native $f < 0$ due to soft/novelty terms and $144{,}185$ with $f \geq 0$, retaining the $-10.0 \leq f < -2.0$ band), with per-run legal counts ranging from $70{,}278$ (35.96%, Run 6) to $80{,}132$ (41.00%, Run 11), a sample mean of $75{,}423.6$, a sample standard deviation (s, $\text{ddof} = 1$) of $2{,}936.55$ (population SD $\sigma = 2{,}785.86$), and a median of $75{,}579.0$ (IQR $= 2{,}908.75$). In Figure 7, Panels A1 (802×440 full $\text{asinh}(f)$ grid over $[-215.0, +1.20]$) and A2 (802×260 bounded linear detail over $[-0.60, +1.16]$ with $727{,}077$ in-window points and $1{,}227{,}483$ conserved in the bottom tail strip) encode occupied generation-by-score bins using logarithmic count shading and per-bin majority legality color (legal $\geq$ penalty); this count aggregation hides minority classifications where legal and penalty rows share a bin and does not render every point separately or equate score sign with legality. Run-maximum native quality-diversity fitness reaches between 1.1089 (Run 3) and 1.1440 (Run 5), with a sample mean of 1.1250, a sample standard deviation of 0.01285 (population SD 0.01220), and a median



Fig. 6. Individual 2000 retained numerical verification records (single assessed export, 1 assessed, 0 qualified, no promoted champion). A–B: three-level plate mesh refinement observations (25,209, 6,803, and 3,166 triangles); both GCI uncertainty estimates are refused ($p = -2.9003$ in A; oscillatory differences in B). C: matched-mesh quadratic tetrahedral solid safety factors (91,008 elements, 416,583 degrees of freedom) comparing independent finite-element solvers under vertex-maximum and vertex-mean recovery. D: planar tip-distance bin maxima (4–25 mm) for raw plate cells versus recovered solid vertex means.

of 1.1207 (IQR = 0.0209). At generation 0, five of the ten runs have non-null legal-population medians (Run 2: −0.2269, Run 4: −0.2070, Run 6: −0.25095, Run 8: −0.2244, Run 11: −0.2620; range −0.2620 to −0.2070) while the remaining five runs have null legal medians at generation 0; across all 801 generations and all ten runs, the minimum legal median is −0.5782 (Run 9, generation 3), and at generation 800 all ten legal-population medians remain negative, ranging from −0.2610 (Run 11) to −0.1933 (Run 4) (mean −0.2272).

Alongside the ten completed 800-generation runs, the repository retains an eleventh staged native-summary collection (Run 12–Run 22, 17,672 generation-by-island summary rows reaching generation 708 with zero internal generation gaps) and nine aborted crescent attempts (7 with partial live histories totaling 216 rows and 2 without CSV rows). Because only the ten completed runs in Cohort 3 retain full 800-generation birth tables, summary-only and aborted runs are reported in separate panels rather than pooled into the 1,954,560-birth denominator. Per-birth Stage-2 CAD assessment records are absent for the 1,954,560 native search rows in Cohort 3, establishing missing retained assessments for those births.

B. Cohort 1: Historical 17-Generation Genealogy and Stage Reversal (Run 1)

In Cohort 1 (Run 1), all 1984 retained births across generations 0–16 are preserved with exact parent links, island assignments, and finite-element evaluation status (1455 FEM-complete records, 529 not-evaluated records, and 446 search-legal births; Figure 7B and Supplement generated/population.svg). Generation 0 contains 63 unevaluated records out of

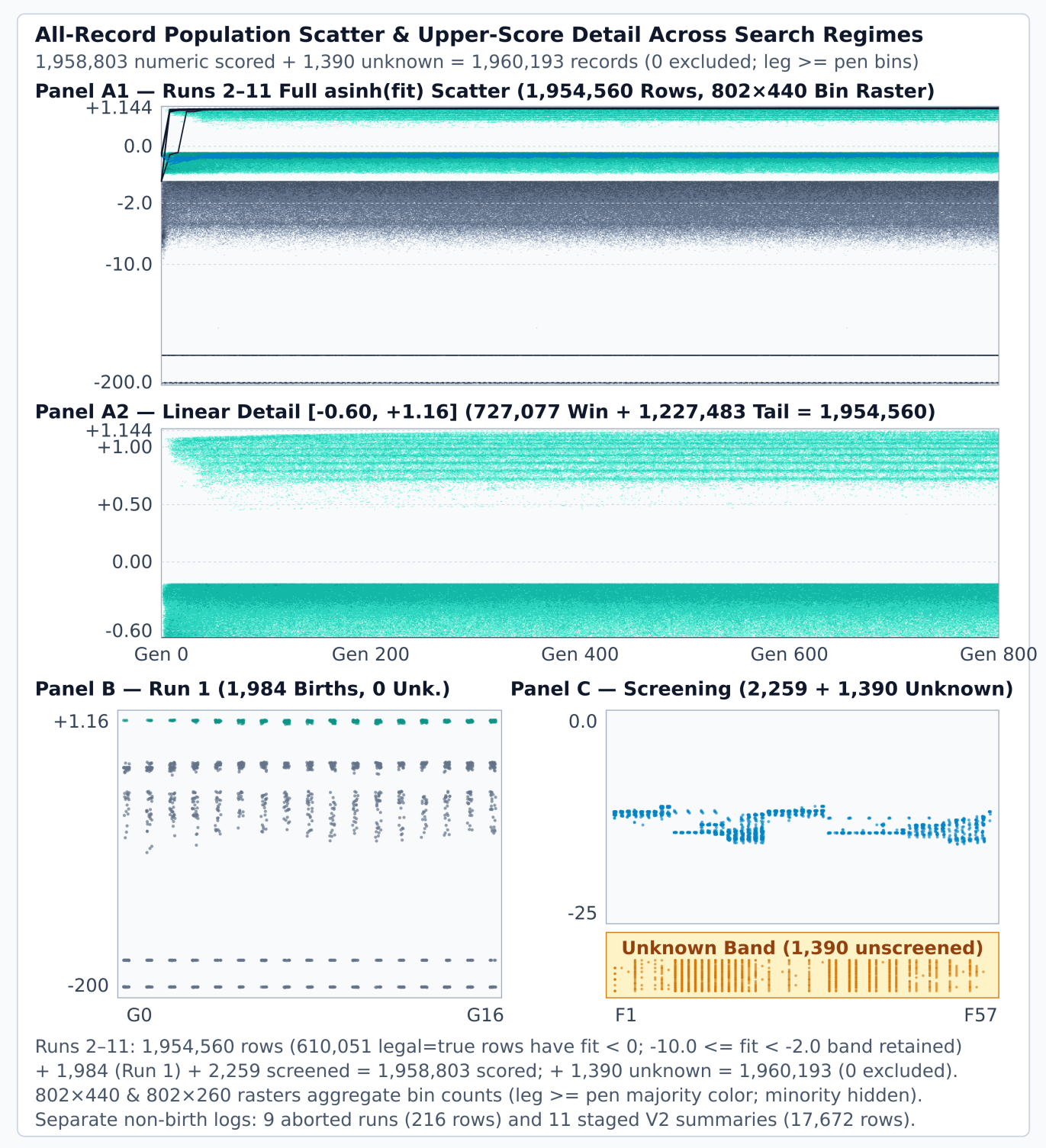


Fig. 7. All-record population scatter and bounded upper-score detail across separate retained search regimes (1,954,560 native Run 2–Run 11 + 1,984 historical Run 1 + 2,259 screened = 1,958,803 numeric scored observations, plus 1,390 unscreened proposals in a separate band = 1,960,193 records, 0 excluded). Panel A1 (802×440 $\operatorname{asinh}(f)$ raster) and Panel A2 (802×260 linear detail on $[-0.60, +1.16]$: 727,077 window + 1,227,483 tail strip = 1,954,560) aggregate per-bin counts with logarithmic shading and per-bin majority legality color (legal $\geq$ penalty, hiding minority overlaps; 610,051 legal rows have $f < 0$ and the $-10.0 \leq f < -2.0$ band is retained), overlaid with per-run cumulative bests ($+1.1089$.. $+1.1440$) and legal medians (G0 non-null -0.2620..-0.2070 across 5 runs; minimum -0.5782; G800 -0.2610..-0.1933).

128, whereas generation 16 contains 21 out of 116, illustrating why pooled score distributions shift as legality-filter pass rates change.

The highest stored birth score in Cohort 1 is 1.15727 for Individual 1095 in generation 9 (display loss -1.15727, raw plate envelope maximum $685.94\,\mathrm{MPa}$). Joining Individual 1095

to the retained strict-assessment summary shows that strict verification was not met (false, failing the regional yield-order sequence threshold). Thus, in this retained historical cohort pairing complete multi-hop genealogy with subsequent strict assessment, the retained historical top-ranked birth failed its later strict assessment.

C. Cohort 2: Intermediate Multi-Stage Screening and Retained Weighted Reporting Profile

Across 24 multi-stage screening run directories, Cohort 2 retains 3,649 birth proposals across 57 round-birth files. Validated one-based candidate indexing joins 2,259 proposals to Stage-1 screening observations, while 1,390 proposals remain in an explicit unscreened observation band (and a zero-based index negative control is verified to reject 2,247 misjoined pairs).

Within the retained weighted reporting profile subset of Cohort 2, six admitted reporting curve files retain fifteen generation summaries (Run 31 through Run 36, archived in Supplement figures/fitness-progress.svg and generated/weighted-runs.svg). Four of the five multi-row series retain their initial archive value across subsequent rounds. In Run 31, the stored archive advances from 74.167488 (Individual 2009 at generation 0) to 74.207299 (Individual 2002 at generations 1–2, display loss 25.792701, 13 favorable legacy comparisons out of 26 overlapping predicates/groups, not qualified), while the third round's best is 73.608446. Because the underlying round-population files and exported CAD geometry for Individual 2002 are absent from the retained directories, these short reporting curves are presented without inventing population uncertainty bands or substituting another cohort's geometry.

D. Separately Assessed Controlled Exports, Automatic Leaderboard, and Battery Reconciliation

In two separately assessed controlled-probe exports (distinct from the evolutionary birth cloud of Run 1), candidate records were evaluated under the 16-gate CAD scorecard:

- **Individual 2001:** Records $W_{\text{CAD}} = -3.878704$, $G_1 = 0.6239029999999999$, raster mass $75.382\,\text{g}$ (3D solid mass $75.1992\,\text{g}$), and passes 11 of 16 legacy gates while failing 5 (certification and refined $260\,\text{N}$ lateral bending, tip-first yield-order ratio ≥ 1.2, $1.82\,\text{N}\cdot\text{m}$ local plate tip torsion, and $1.82\,\text{N}\cdot\text{m}$ quadratic solid tip submodel).
- **Individual 2007:** Records $W_{\text{CAD}} = -3.964924$, $G_1 = 0.6216340416666666$, raster mass $75.642\,\text{g}$ (3D solid mass $75.5049\,\text{g}$), and likewise passes 11 of 16 legacy gates while failing the same 5 gates.

Both controlled exports failed grouped strict requirement verification.

In the fixed 23-subject historical CAD battery ($23\times18 = 414$ cells; preserved in the linked Supplement generated/battery-matrix.svg and Section X), 221 stored passes, 81 failures, and 112 blocked records are reconciled, with 92 passes attributable to four shared software-control columns and 20 passes in the quadratic-solid tip column recording file completeness rather than favorable strength. Separately, the automatic certification leaderboard contains a single assessed export (Individual 2000; 1 assessed candidate, 0 qualified, no promoted champion), which is not qualified.

E. Research Questions Answered and Open

Our empirical findings answer the study's central research question through bounded, reproducible evidence: (1) the 129-field executable genome and island variation operators generate inspectable gene-coordinate records and recorded population changes across 1,954,560 evaluated births in Cohort 3 and 1,984 genealogically linked births in Cohort 1; (2) the retained historical top-ranked birth (Individual 1095 in Cohort 1) failed its later strict assessment, and separately assessed exports (Individual 2000, Individual 2001, Individual 2002, and Individual 2007) likewise fail strict multi-gate qualification; and (3) matched-mesh quadratic solid solvers agree within 0.05% while three-level plate GCI estimates must be refused when observed orders are negative or oscillatory. Measuring formal structural diversity, conducting representation ablations, fabricating physical specimens, and performing cyclic fatigue testing remain open prospective work.

IX. Agents and the Next Level of Genome Design

A. From Parameter Search to Typed Representation Proposals

Searching a fixed vector and changing the representation are different activities: the first explores values under a stable decoder, whereas the second changes which objects can be expressed and how coordinates relate to observations. Agents could assist representation research by proposing typed schemas, explaining dependencies, locating unsupported assumptions, and drafting testable software changes. This is a prospective capability, not an outcome established by the retained runs.

A reviewable representation proposal should state its purpose, schema version, typed fields, units, decoder dependencies, serialization rules, migration behavior, and test oracle, separating intended effects from observed decoder and solver outputs. Only decoder outputs establish what the code produced, and only numerical or physical observations establish model or manufactured behavior.

B. Independent Evaluation and Failure Modes

The evaluator must remain independently controlled relative to the proposer. An agent that changes a representation must not silently alter acceptance criteria, nor may a reporting adapter redefine missing data as favorable. Separate source authorship, separate runtime execution, and separate model families test different dimensions of independence and should be named explicitly.

Tractable failure modes include unit mismatches, inactive parameters, decoder ambiguities, invalid geometry, incompatible migrations, and evaluator exploitation—producing cases that score well under a proxy while violating the assumptions that made the proxy meaningful. These risks should be measured through invalid-proposal rates, review outcomes, reproducibility, and oracle agreement rather than prose fluency.

C. Prospective Benign Research and Human Responsibility

A future controlled study on benign reference CAD tasks could compare human-authored and agent-assisted schema proposals under fixed review criteria and evaluator versions, retaining rejected proposals and measuring semantic clarity, decoder reliability, independent reconstruction, and reviewer effort. Expanding the hypotheses considered does not remove human responsibility for problem definition, model selection, evidence interpretation, and deployment: a human author must approve the manuscript's claims, while numerical tools and physical measurements constrain what can be trusted.

X. Reproducibility, Supplement Crosswalk, and Publication Provenance

A. Four Distinct Reproduction Claims

This manuscript reconstructs documentation projections and population censuses from retained repository records; it does not rerun finite-element solvers or validate physical hardware. The standalone Rust verification package reads the frozen battery and science catalogs, three-cohort population ledgers, identity registries, and compressed generation-800 archives in read-only check mode without invoking CAD or FEM kernels. Chart reconstruction, solver replay, software-control execution, and physical replication establish different scientific claims and are kept strictly separate.

B. Integer-Only Reader Identities and Linked Technical Supplement

To keep the main narrative readable with integer-only individual and run identifiers while preserving byte-level auditability, the linked Technical Supplement (docs/paper/CROSSWALK_SUPPLEMENT.md and docs/paper/supplement.tex, governed by docs/paper/data/public_ids.v1.json and docs/paper/publication.v2.json) maps every public identifier to its internal repository alias, producer revision, file path, and SHA-256 digest:

- **Individual 2000:** Single assessed export in the automatic certification leaderboard (1 assessed candidate, 0 qualified, no promoted champion; STEP solid, 90,000-triangle STL, and 25,209-triangle plate mesh across seven arrays; continuous CAD score $W_{\text{CAD}} = -5.570408$ via the failed-branch formula $-1.0-4.619108+0.10s_{\text{mass}}$ across refined tip torsion 0.4710, refined plate bending 0.3531, and quadratic solid tip 0.3307, versus pass-branch quality 1.456983, with $10/16$ legacy gates and $2/10$ requirement groups met). Separate from the 23-subject historical battery; the 1,954,560 native search births in Cohort 3 lack retained Stage-2 CAD assessments.
- **Individual 1095 (Cohort 1 Run 1, generation 9):** Highest native search birth ($f = 1.15727$) across $1{,}984$ retained births and 19 binary event frames (2,207 events, 2,176 island-slot memberships); fails strict failure-sequence verification.
- **Individual 2001, Individual 2007, and Individual 2008:** Separately assessed controlled exports. Individual 2001 records $W_{\text{CAD}} = -3.878704$, $G_1 = 0.623903$, and mass $75.382\,\text{g}$ ($75.1992\,\text{g}$ 3D solid), passing 11 of 16 legacy gates and failing 5; Individual 2007 records $W_{\text{CAD}} = -3.964924$, $G_1 = 0.621634$, and mass $75.642\,\text{g}$ ($75.5049\,\text{g}$ 3D solid), passing 11 of 16 legacy gates and failing the same 5 gates. Historical 16-gate probe Individual 2008 records $W_{\text{CAD}} = +1.7790$, $G_1 = 0.878210$, and mass $75.568\,\text{g}$ ($75.323\,\text{g}$ 3D solid) while failing grouped fatigue and edge-profile predicates.
- **Individual 2002 and Individual 2009 (Run 31 in Run 31–Run 36):** Across generations $0, 1, 2$ of Run 31, cumulative archive scores are 74.167488 (Individual 2009, $101.108\,\text{g}$), 74.207299 (Individual 2002, display loss 25.792701, $101.139\,\text{g}$, $13/26$ legacy flags), and 74.207299, while round bests are 74.167488, 74.207299, and 73.608446.
- **Cohort 3 Ten-Run Archive (Run 2–Run 11) and Supplement 414-Cell Battery Matrix:** Each run's $195{,}456$-row evaluation table is partitioned into twenty $\leq 8\,\text{MiB}$ compressed segments (200 segments total, compressing 3,811,896,749 raw bytes into 494,843,141 stored bytes with verified 100% SHA-256 round-trip identity; earlier staging ordinals 37..46 map one-to-one to Run 2–Run 11). The exhaustive 23-subject by 18-column historical battery matrix (414 cells: 221 passes, 81 failures, 112 blocked records), export-identity diagram, and contributor-support plate audit are preserved in the linked Supplement.

C. Deterministic Verification Gate

The repository verification gate executes two read-only checks from the repository root:

```
cargo run --release --locked --manifest-
path \
  crates/paper-artifacts/Cargo.toml \
  --example publication_v2_audit
make -C docs/paper check
```

The first command verifies all metadata contracts, decodes the binary event frames of Run 1, verifies the 57-file one-based join and zero-based negative control of Cohort 2, decompresses and checks the 200 compressed segments (3,811,896,749 raw bytes) and scatter-cache bindings of Cohort 3, and confirms byte-for-byte freshness of all generated outputs without writing files. The second command renders the SVG figures, compiles `main.pdf`, verifies zero undefined references or overfull boxes, and checks the 15–20 page budget.

XI. Discussion

A. What a Scalar Cannot Say Alone

A scalar expresses preference over its supplied observations. Equal totals can hide different component profiles, and an increase can come from a changed proxy rather than physical behavior. Retaining the component vector and model identity exposes these differences. A decreasing loss within a stable definition guarantees neither improvement of every component nor an optimum as compute increases. The historical and weighted scores examined here cannot supply one common trajectory.

B. Claims About Strength, Mass, and Practical Use

The retained data include modeled mass and mechanical responses. These are valuable observations, but their practical interpretation remains bounded by model assumptions, manufacturing state and missing validation. A low CAD mass does not establish the mass of a finished assembly, and a modeled safety factor does not establish reliability under an unvalidated use scenario. The historical application should not be used to imply operational suitability from a selected numerical example.

Transfer also depends on the represented family and material state. A titanium label does not encode process history, orientation, porosity, surface condition or inspection outcome. A new domain requires its own model assumptions and measured evidence.

C. Manufacturing Freedom and Precision

Metal additive manufacturing can make geometries that are difficult to create through conventional routes. Geometric freedom, however, is different from dimensional precision. A digital model can specify arbitrarily many decimal places without the manufactured object achieving those dimensions. Thermal behavior, surface formation, postprocessing and measurement all influence the relationship between CAD and part.

NIST's geometric test-artifact work provides a concrete reason to measure process capability rather than infer it from the digital model [10]. The paper therefore treats unprecedented tolerance as a hypothesis requiring metrology, not a conclusion from optimization. A future study would need defined dimensional characteristics, calibrated measurement methods, uncertainty estimates and a comparison baseline. Nominal CAD fidelity alone cannot answer that question.

D. Threats to Validity

This is a retrospective single-repository study. The retained cohorts were selected by prior workflows and are not random samples. Some source files are unavailable. Historical model and objective revisions limit cross-wave comparison. Shared observations create correlated outcomes, and a large number of labels can exaggerate apparent corroboration. The new publication code verifies projections of retained bytes but cannot establish the truth of every original computation.

Independent reconstruction, controlled agent-assisted representation experiments and physical manufacturing studies remain absent. Author reporting checks and adversarial review do not replace those experiments. The contribution is a retrospective system study of retained rankings and numerical-assessment evidence, supported by a reproducible reporting package.

E. Research Questions Opened by the Results

The separation between search ranking and later assessment motivates a controlled study on a benign reference problem: how often do rankings survive independently specified assessment, and which differences explain reversals? Reporting research could measure reconstruction success and detection of deliberately introduced provenance defects. Agent research could compare typed representation proposals under fixed independent criteria, retaining invalid and rejected attempts. These are prospective experiments, not additional results of this study.

The linked architecture makes these questions inspectable: it retains representation identity, multiple model scopes, solver records and failed interpretations. Its long-term value is to reduce the cost of exploring design alternatives while retaining enough evidence to determine which apparent gains survive numerical and physical scrutiny.

XII. Conclusion: From Unexpected Geometry to Valuable Products

The future of computational engineering is the ability to formulate and explore geometric families, explain their tradeoffs, and connect every conclusion to verifiable evidence. Jarambit's engineering depth becomes legible when 16 diagnostic predicates, ten overlapping requirement groups, 18 heterogeneous evaluator columns, and 178 assurance definitions are explained through their physical and numerical roles rather than advertised as one aggregate test count.

The retained evidence supports a bounded conclusion: the highest historical search rank failed later strict assessment, and the separately assessed automatic export reports $< 0.05\%$ matched solid-solver agreement while its three-level plate GCI estimates are refused. Together, these cases show why search ranking, solver-implementation parity, and discretization convergence require separate interpretations. Reconstructing the three-cohort population archive and historical battery within their declared scopes establishes neither a qualified product, global optimality, agent-discovered representations, unprecedented manufacturing tolerance, nor physical reliability.

Joining representation research with additive manufacturing and metrology opens prospective opportunities across aerospace fixtures, robotics, thermal management, medical-device research, scientific instruments, and energy systems, each requiring domain-specific models, physical measurements, and qualification. The path from a surprising digital form to a trusted engineering product runs through metrology, uncertainty quantification, independent scrutiny, and reproducible evidence.

Acknowledgment and Assistance Disclosure

This draft was prepared with AI assistance for source analysis, manuscript organization and prose, documentation tooling, and author adversarial review. Human author approval, contribution verification, independent scientific review, and venue-specific disclosure remain required before submission. No institutional affiliation or funding is asserted.

References

[1] J. Taylor and repository contributors, "Jarambit source and retained evidence," revision `009f5ac520fa3afe8ed8025230eef604dbb7f922`, 2026. [Online]. Available: https://github.com/neverhuman/jarambit/tree/009f5ac520fa3afe8ed8025230eef604dbb7f922.

[2] R. Wu, C. Xiao, and C. Zheng, "DeepCAD: A deep generative network for computer-aided design models," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021, pp. 6772–6782. [Online]. Available: https://www.cs.columbia.edu/cg/deepcad/.

[3] M. S. Khan, S. Sinha, T. U. Sheikh, D. Stricker, S. A. Ali, and M. Z. Afzal, "Text2CAD: Generating sequential CAD designs from beginner-to-expert level text prompts," in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 7552–7579, doi: 10.52202/079017-0242. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2024/hash/0e5b96f97c1813bb75f6c28532c2ecc7-Abstract-Conference.html.

[4] J. D. Lohn, D. S. Linden, G. S. Hornby, W. F. Kraus, A. Rodríguez-Arroyo, and S. E. Seufert, "Evolutionary design of an X-band antenna for NASA's Space Technology 5 mission," NASA Technical Reports Server, document 20030067398, 2003, conference preprint. [Online]. Available: https://ntrs.nasa.gov/citations/20030067398.

[5] J.-B. Mouret and J. Clune, "Illuminating search spaces by mapping elites," arXiv:1504.04909, 2015. [Online]. Available: https://arxiv.org/abs/1504.04909.

[6] B. Romera-Paredes *et al.*, "Mathematical discoveries from program search with large language models," *Nature*, vol. 625, pp. 468–475, 2024, doi: 10.1038/s41586-023-06924-6. [Online]. Available: https://doi.org/10.1038/s41586-023-06924-6.

[7] NASA, *Standard for Models and Simulations*, NASA-STD-7009B, Mar. 5, 2024. [Online]. Available: https://standards.nasa.gov/standard/NASA/NASA-STD-7009. Cited for credibility context; compliance not claimed.

[8] T. Gustafsson and G. D. McBain, "scikit-fem: A Python package for finite element assembly," *J. Open Source Softw.*, vol. 5, no. 52, p. 2369, 2020, doi: 10.21105/joss.02369. [Online]. Available: https://joss.theoj.org/papers/10.21105/joss.02369.

[9] I. A. Baratta *et al.*, "DOLFINx: The next generation FEniCS problem solving environment," preprint, version 1, 2023, doi: 10.5281/zenodo.10447666. [Online]. Available: https://zenodo.org/records/10447666.

[10] S. Moylan, J. Slotwinski, A. Cooke, K. Jurrens, and M. A. Donmez, "An additive manufacturing test artifact," *J. Res. Natl. Inst. Stand. Technol.*, vol. 119, pp. 429–459, 2014, doi: 10.6028/jres.119.017. [Online]. Available: https://www.nist.gov/publications/additive-manufacturing-test-artifact.

[11] SciPy developers, "scipy.sparse.linalg.eigsh," API reference, accessed Oct. 19, 2026. [Online]. Available: https://docs.scipy.org/doc/scipy/reference/generated/scipy.sparse.linalg.eigsh.html. Cited for the symmetric/Hermitian-operator contract; the archived A4 SciPy version is not recorded.

[12] P. J. Roache, "Perspective: A method for uniform reporting of grid refinement studies," *J. Fluids Eng.*, vol. 116, no. 3, pp. 405–413, 1994, doi: 10.1115/1.2910291. [Online]. Available: https://doi.org/10.1115/1.2910291.